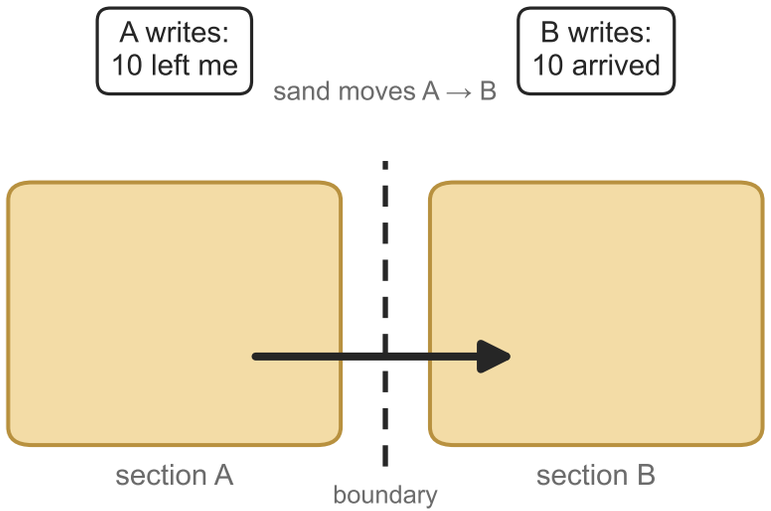
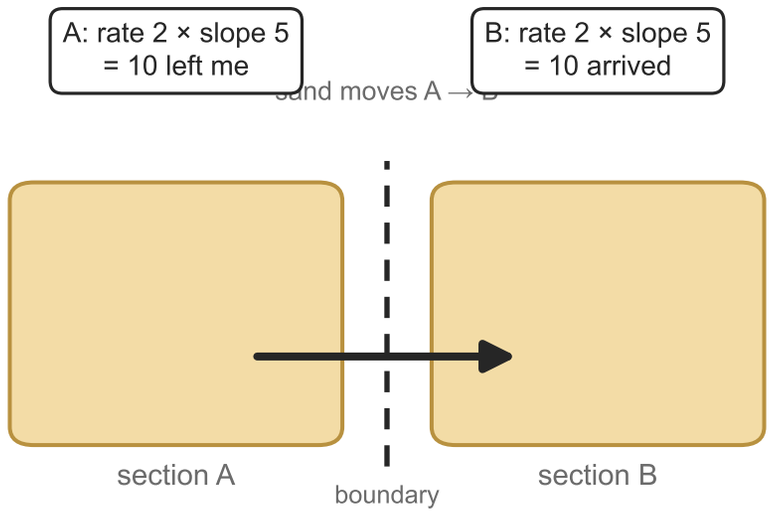
The bookkeeping problem, without any model (illustrative numbers)

1 What should happen: sand that leaves A arrives in B, and both write down the same amount



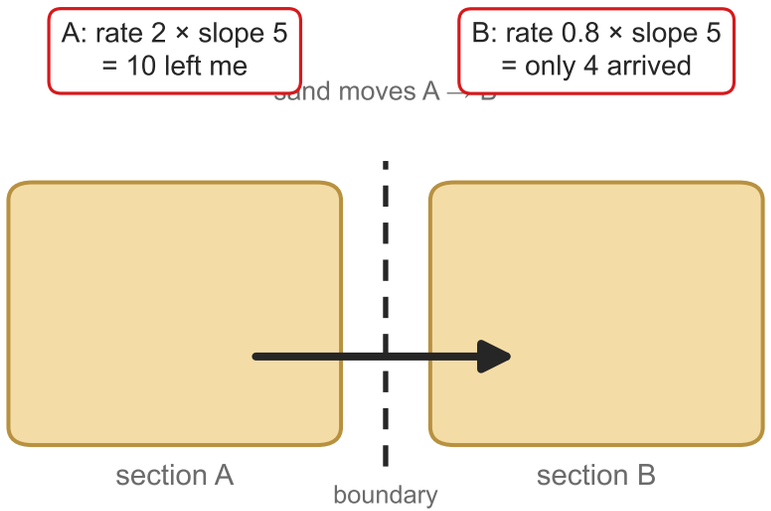
−10 + 10 = 0: nothing made or lost

2 BRIE: each side works out the amount itself: its OWN rate × the slope between them



same rate on both sides: the books agree

3 If the two sides' rates differ, they write down different amounts for the same handover



−10 + 4 = −6: 6 vanished on the way